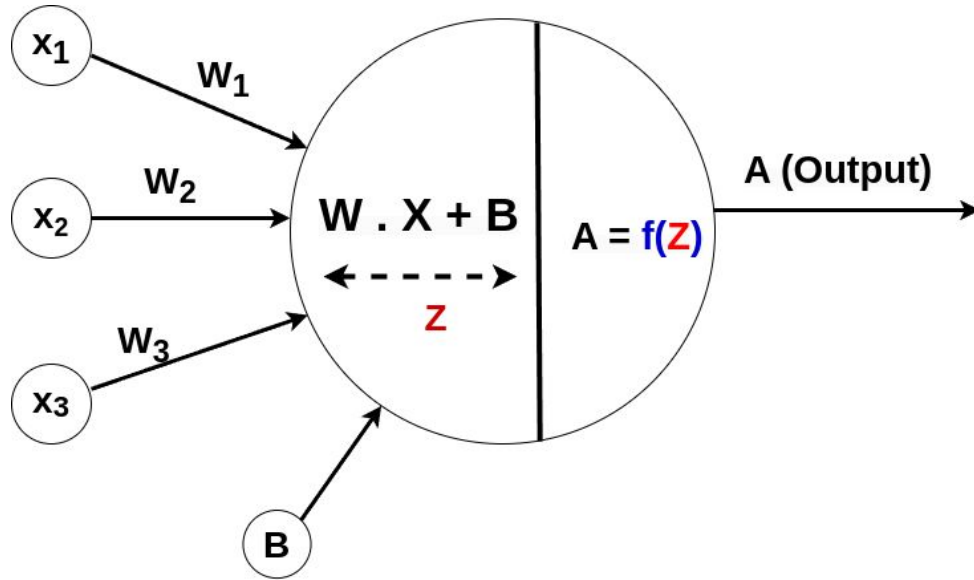


Intro to Neural Network

One neuron



X = input

W = weights

B = Bias

f = Non - Linear Function

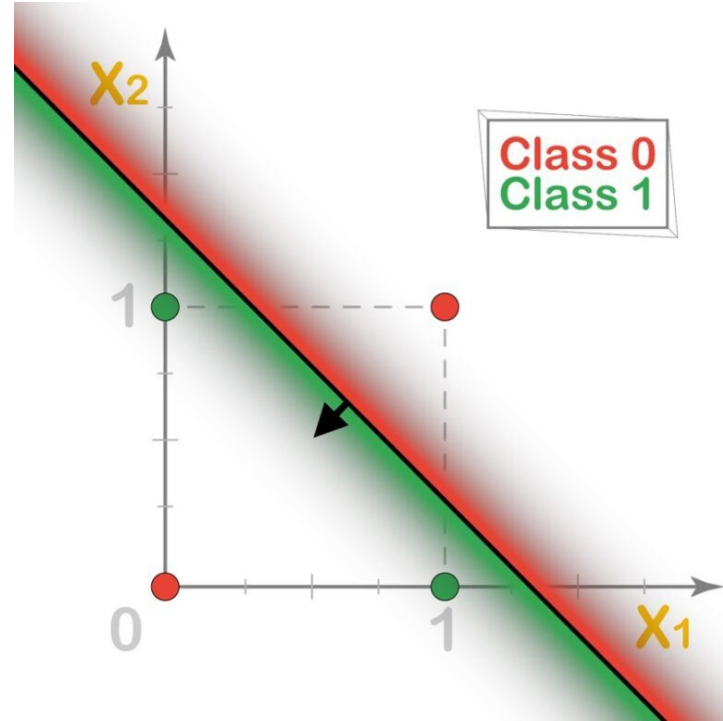
A = Output

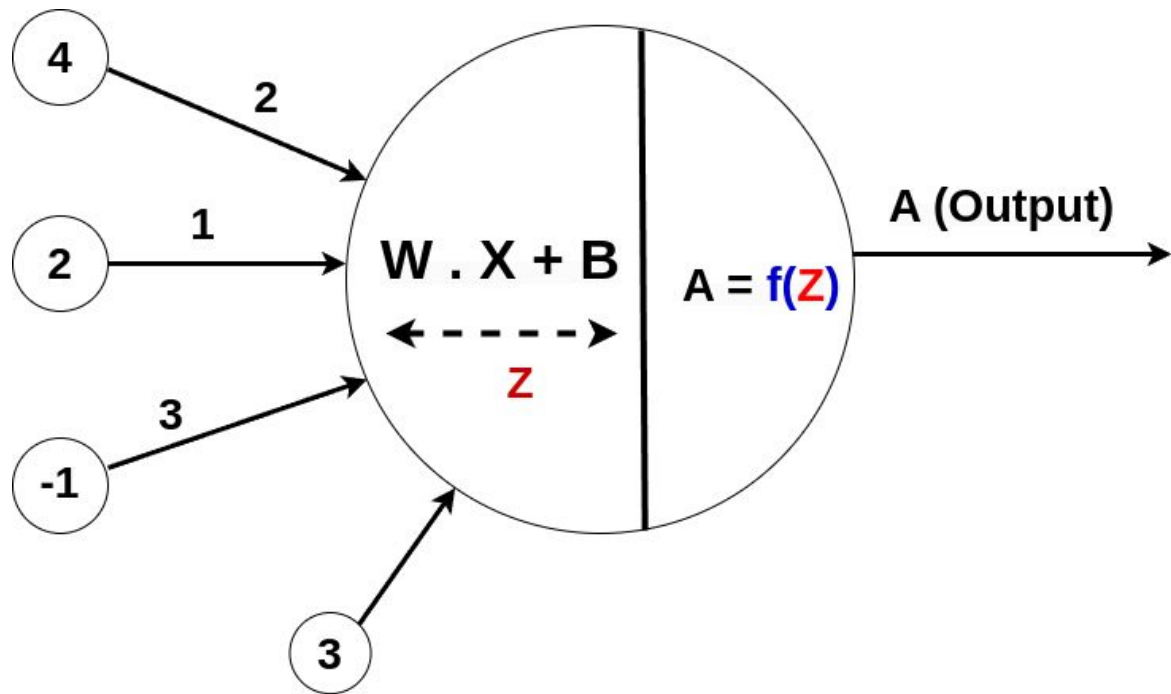
$$A = f [W_1 X_1 + W_2 X_2 + W_3 X_3 + B]$$

Why introduce non-linearity

Can a linear line separate green and

Red lines ? **NO**



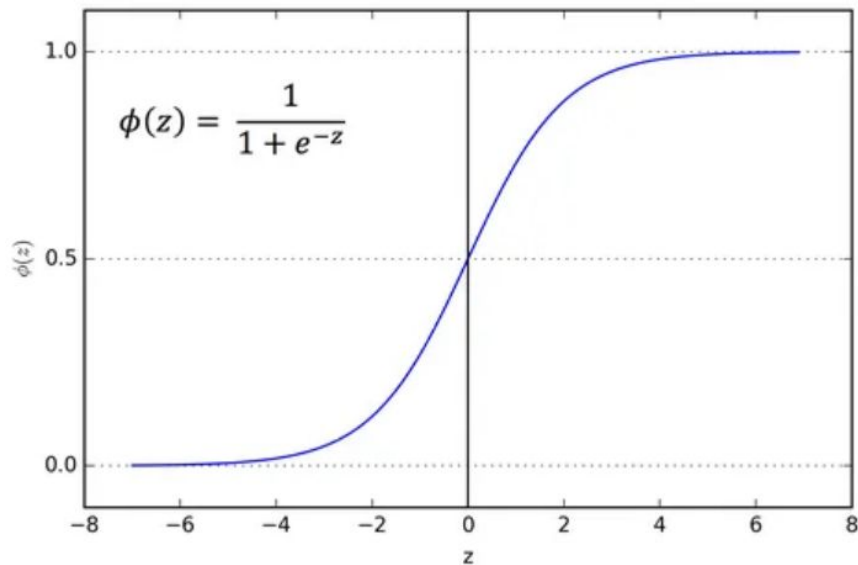


Activation Function

- Sigmoid
- ReLU
- Tanh
- Softmax

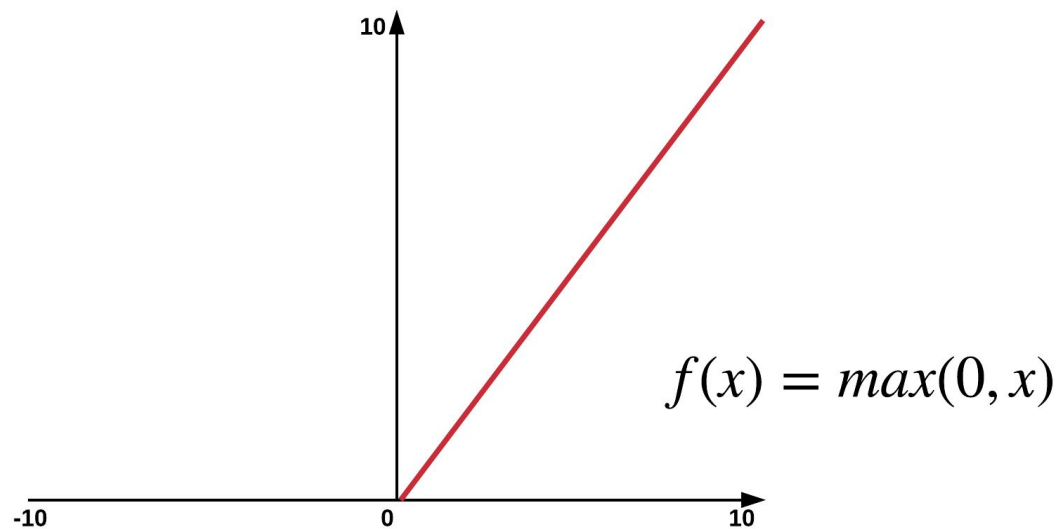
Sigmoid Function

- Commonly used on the **Output layer**
- Used in Binary Classification.



ReLU

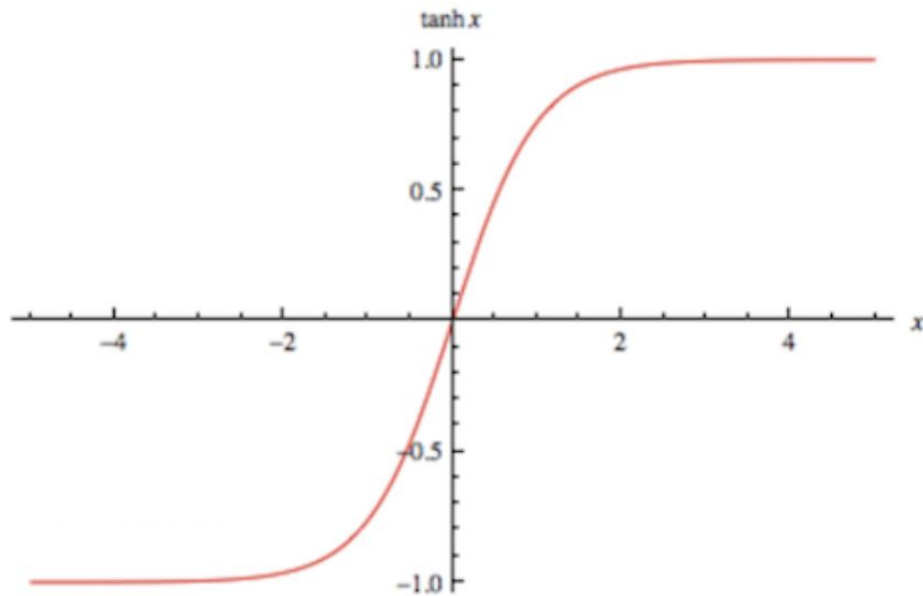
Commonly used in Hidden Layers



Tanh

Not used commonly now.

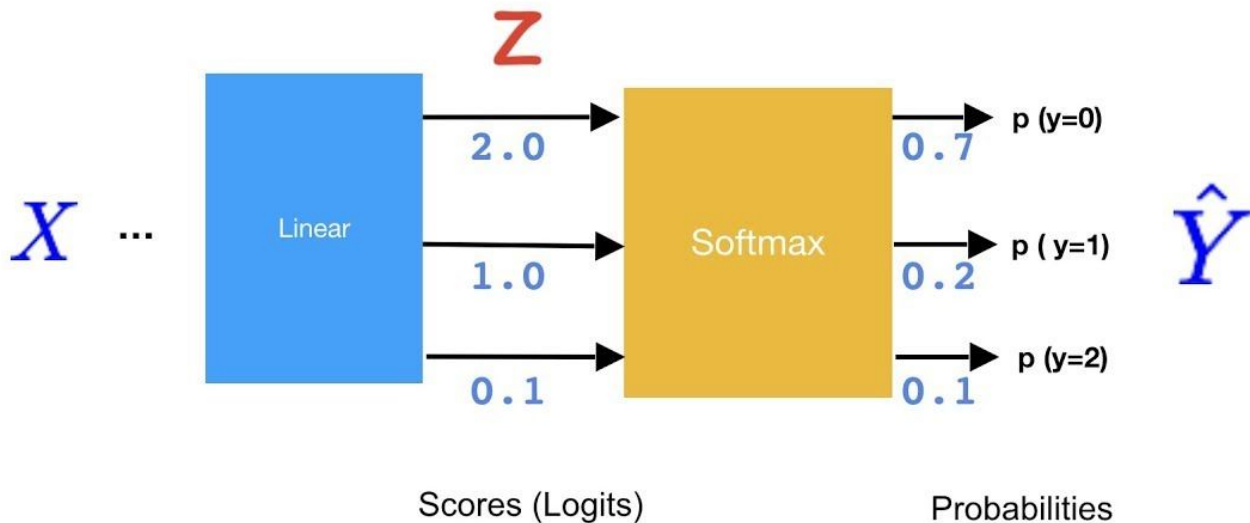
Used with traditional RNN



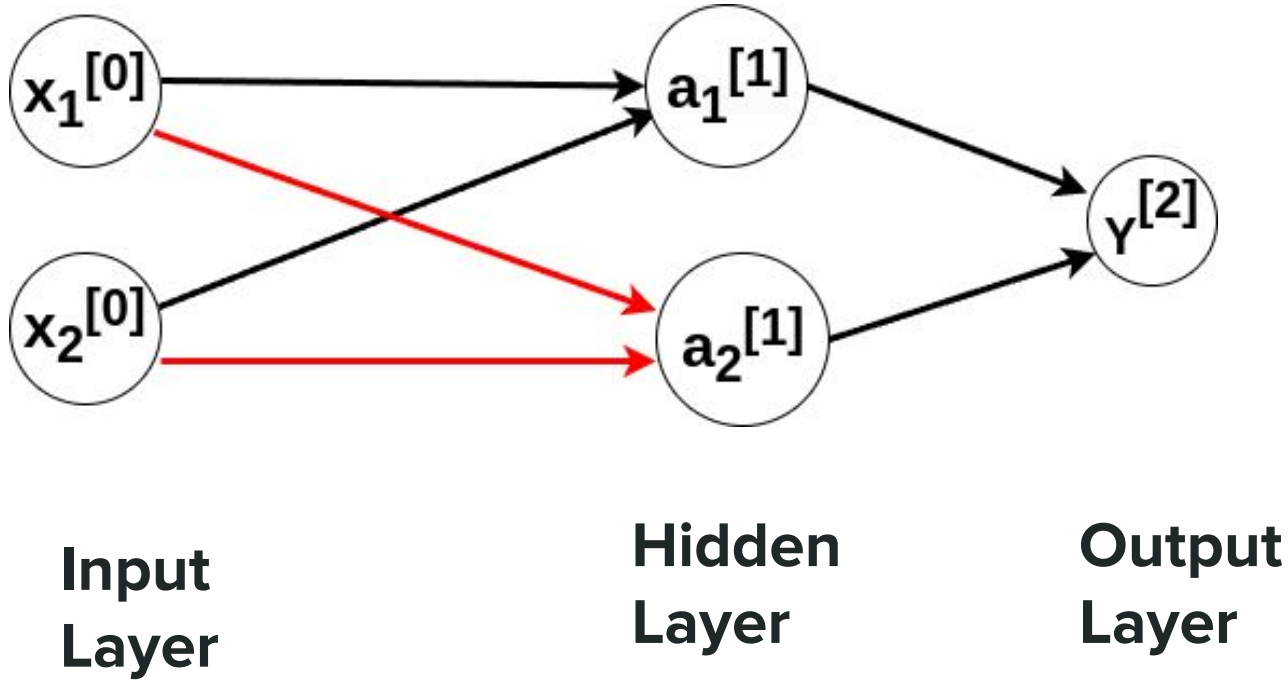
Softmax

Used as Output Layer
In **MultiClass Classification**

$$\sigma(\mathbf{z})_j = \frac{e^{z_j}}{\sum_{k=1}^K e^{z_k}} \quad \text{for } j = 1, \dots, K.$$



Multilayer Perceptron (MLP)



Forward Propagation

$$z(0) = W(0)x + b(0)$$

$$a(0) = f[z(0)]$$

$$z(1) = W(1)a(0) + b(1)$$

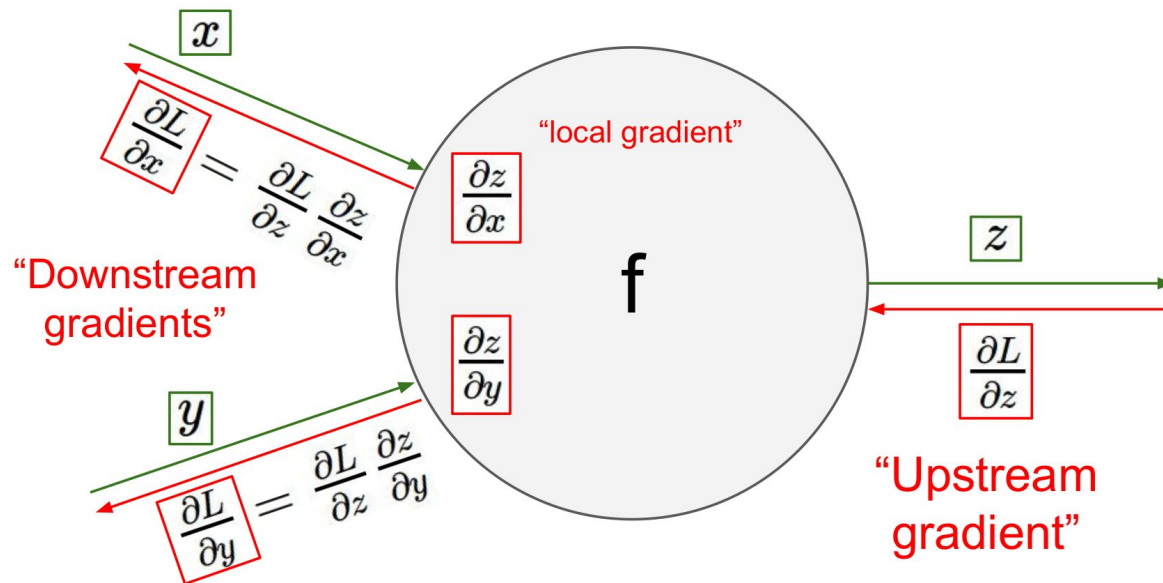
$$a(1) = f[z(1)]$$

Binary Cross entropy loss (Binary Classification)

$$\text{Loss} = -\frac{1}{\text{output size}} \sum_{i=1}^{\text{output size}} y_i \cdot \log \hat{y}_i + (1 - y_i) \cdot \log (1 - \hat{y}_i)$$

Type Of Task	Loss Function	Output Neuron Count	Activation Function
Binary Classification	BinaryCrossEntropy	1 neuron	Sigmoid
MultiClass Classification	CategoricalCrossEntropy	Number of Outputs	Softmax
Regression	Mean Squared Error		

Backward Propagation



One Iteration

During Training Phase,

First calculate the prediction using forward propagation.

Calculate the loss comparing the predicted value and the actual value

Then backpropagate the error. Keep adjusting the weight.

One iteration COMPLETED. Now lets do this same thing again for the next iteration.

Optimizers

1. Gradient Descent
2. Stochastic Gradient Descent (SGD)
3. Mini Batch Stochastic Gradient Descent (MB-SGD)
4. SGD with momentum
5. Nesterov Accelerated Gradient (NAG)
6. Adaptive Gradient (AdaGrad)
7. AdaDelta
8. RMSprop
9. Adam

[Website Link](#)

Epoch vs Batch (size)

Epoch = No. of iteration of the entire training set

Batch (Size) = No. of data points or samples updated for each weight updates.

Do Note, there is number of batches which is different

Example Calculation

There is a dataset with 100 samples. The batch size is 25. And epoch is 10

Show the calculation.

Hyper Parameters to Consider

- Number of hidden layers
- Number of neurons per layer
- Activation function per layer
- Optimizer
- Learning Rate
- Batch size and epoch
- Dropout
- Early Stopping
- Regularization
- Batch Normalization

Types of Neural Network

MLP → Tabular Data

RNN (Recurrent Neural Network) → Sequence data like text data

CNN (Convolutional Neural Network) → Image dataset

GAN (Generative Adversarial Network) → Fake image detection

Libraries to Implement Neural Network

- Tensorflow with Keras → Commonly CV
- Pytorch → Commonly NLP